MOH MAYA

World & Engine Definition
Technical Design Document — V1.0

Defining the simulated world and the engines that make it alive.

Project: Moh Maya
Version: 1.0 — Foundational Technical Model
Created by: Victus

INDEX

1. Purpose

2. What Is the World?

3. World Principles

4. World State

5. World Constraints

6. Constraint Pressure

7. Ticks & Time

8. Engine Model

9. World Engine

10. Simulation Engine

11. Character Engine

12. Knowledge & Memory Engine

13. Relationship Engine

14. Health & Life Engine

15. Economy & Currency Engine

16. Government & Society Engine

17. Event Engine

18. LLM Layer

19. God Layer

20. Engine Interaction

21. Example: Drought

22. Hard Rules

23. V1 Scope

24. North Star

1. Purpose

This document defines what the simulated world of **Moh Maya** is and the major engines responsible for keeping that world alive.

It translates the foundational Moh Maya Bible into a first technical model. It defines responsibilities and boundaries, not programming languages, frameworks, databases, or specific LLM vendors. The Bible establishes that the world exists independently of the player, that the player only knows what they know, that relationships exist without the player, and that time must have meaningful consequences. ■filecite■turn0file0■

2. What Is the World?

The Moh Maya world is a small, persistent physical place — initially a town or small city — populated by people, governed by rules, constrained by resources, and continuously changing through time.

The world should approximate the real world of today closely enough that its rules feel intuitive. This does not require full Newtonian physics. Instead, Moh Maya has **world physics**: physical, biological, economic, social, environmental, and institutional rules that determine what is possible and what pressures the world.

The player is one person inside this world. The world does not exist solely for the player and does not stop when the player is absent.

The world contains

• Geography and locations

• Population

• Homes, workplaces, businesses and public places

• Roads and infrastructure

• Natural and environmental conditions

• Resources and resource limits

• Time and schedules

• Health and biological processes

• Money, employment and economic activity

• Relationships and social structures

• Government and institutions

• Events and accumulated history

3. World Principles

The world is persistent. It continues to exist and change beyond the player's direct interaction.

The world is causal. Events should have understandable causes in state, constraints, decisions, or deliberate interventions.

The world has limits. Resources, capacity, time, money and physical locations matter.

The world is imperfectly known. The simulation can know facts that the player and characters do not know.

The world is autonomous. Characters and institutions can act without waiting for the player.

The world changes with time. Time produces meaningful changes in people, resources, relationships and institutions.

The world produces consequences. Events change state and can create further events.

The world is not scripted around the player. The player influences the world but is not its sole source of activity.

4. World State

At any moment the world has an authoritative state. It is the foundation from which the simulation determines what can happen next.

• Current date and time

• Locations and presence

• Population and individual states

• Available resources

• Infrastructure and capacity

• Environmental conditions

• Economic conditions

• Government and institutional state

• Social relationships and groups

• Active events and unresolved situations

• Historical facts and consequences

5. World Constraints

A constraint represents a limit, capacity, dependency, or condition that influences behavior.

• Finite water

• Hospital bed capacity

• Household money

• Human time and energy

• Business inventory

• Road capacity

• Housing supply

• Government resources

• Physical location constraints

• Biological time requirements

Constraints turn the world from a narrative backdrop into a system with consequences.

6. Constraint Pressure

The world should not only react when a constraint is broken. It should begin behaving differently as the constraint approaches its limits.

Normal → Increasing pressure → Warning → Critical → Constraint violation / crisis

For example, a reservoir does not need to reach zero before water stress begins affecting prices, decisions, government responses, health, relationships and other systems.

Constraint pressure is one of the mechanisms by which the world can naturally produce events instead of relying only on random event generation.

7. Ticks & Time

Time advances through discrete **ticks**. A tick is the heartbeat of the simulation. The exact real-world duration represented by one tick is a later decision.

Conceptually a tick advances time, updates time-dependent state, evaluates constraints, processes autonomous activity, processes events, applies consequences, records important changes, and produces the next valid world state.

The tick must be able to advance the world even when the player is doing nothing.

8. Engine Model

Moh Maya should not be one giant AI brain. The world is composed of engines with clear responsibilities.

Engine	Question it answers
World Engine	What exists and what is the current state?
Simulation Engine	What changes when time and rules are applied?
Character Engine	Given circumstances, what might a person do?
Knowledge & Memory	Who knows, remembers, believes or experienced what?
Relationship Engine	How do connections between people change?
Health & Life	How do biological and life conditions evolve?
Economy & Currency	How do money, work, resources and financial pressure move?
Government & Society	How do institutions, laws and collective decisions operate?
Event Engine	What events arise, occur and continue?
LLM Layer	How are nuanced human decisions and interactions reasoned about?
God Layer	What deliberate conditions or events are introduced by developers?

9. World Engine

The World Engine is the authoritative representation of what exists.

• Maintain world state

• Maintain locations and physical presence

• Maintain environment and resources

• Maintain infrastructure and capacity

• Expose authoritative world facts

• Reject impossible state changes

The World Engine is the authority on reality. An LLM may suggest an action or response, but it cannot simply invent a world state that contradicts authoritative state.

10. Simulation Engine

The Simulation Engine moves the world from one valid state to the next.

• Advance ticks

• Apply world rules

• Process time-dependent changes

• Evaluate constraints and pressure

• Resolve valid state transitions

• Coordinate engines

• Trigger or update events

• Record significant changes

This is the machinery that turns a static world model into a living world.

11. Character Engine

The Character Engine represents autonomous people as participants in the world.

A character has circumstances, personality, goals, knowledge, memories, relationships and limitations. The engine determines when a character needs to make a decision and provides the context needed for that decision.

• Going to work

• Calling a friend

• Avoiding someone

• Buying something

• Looking for a job

• Helping family

• Attending an event

• Responding to a crisis

• Changing plans when circumstances change

Characters must be able to act when the player is absent.

12. Knowledge & Memory Engine

This engine enforces the information boundary established by the Bible.

• Objective truth

• What a character knows

• What a character remembers

• What a character believes, including incorrect beliefs

• What the player experienced

• What the player learned from others

The simulation can know more than the player. Hidden world truth must not automatically become player knowledge.

Dynamically established history becomes persistent continuity. This follows the Bible's rule that established facts become canon. ■filecite■turn0file0■

13. Relationship Engine

Relationships are multidimensional rather than a single friendship score. Trust, affection, resentment, dependence, loyalty, fear, respect and conflict may coexist.

Relationships evolve through arguments, favors, betrayal, support, absence, misunderstandings, shared experiences, loss, achievements and time.

Characters can form and change relationships independently of the player, creating hidden social stories. ■filecite■turn0file0■

14. Health & Life Engine

This engine models biological and life-state changes.

• Age

• Basic health

• Illness and recovery

• Energy and rest

• Major life stages

• Other time-dependent biological conditions

15. Economy & Currency Engine

The economy supports believable life decisions rather than becoming a full national economic simulator.

• Income

• Employment and wages

• Expenses

• Savings

• Debt

• Prices

• Business revenue and costs

• Resource scarcity

• Financial pressure

This matches the Bible's intentionally simple V1 economic scope. ■filecite■turn0file0■

16. Government & Society Engine

This engine makes institutions part of the living world.

• Government state

• Laws

• Taxes

• Public services

• Elections

• Candidates and political positions

• Basic public opinion

• Government response to crises

The goal is a functioning social system, not a separate political strategy game.

17. Event Engine

Events have causes, participants, state and consequences.

Emergent: arise from world conditions and character behavior.

Scheduled: arise from calendars and recurring systems.

God: deliberately introduced by developers.

> *World condition → pressure → event → character response → consequence → new world condition*

The Event Engine should connect events to world state rather than produce disconnected random stories.

18. LLM Layer

The LLM is not the simulation. It is a reasoning and interaction layer inside the simulation.

The simulation establishes what is true and what is possible. The LLM can reason about how a character interprets circumstances, chooses among plausible actions, communicates, or responds to complex human situations.

• Nuanced character decisions

• Conversation

• Human interpretation

• Social reasoning

• Consistent character responses

• Relevant history generation

• Plausible human reactions

• Narrative expression of events

The LLM must not be the authoritative clock, resource counter, location database, or unrestricted event generator.

19. God Layer

The God Layer represents deliberate developer intervention.

• Starting conditions

• Major external events

• Disasters

• New laws or political changes

• Economic shocks

• Special scenarios

• Experimental conditions

God should create circumstances, not necessarily outcomes.

For example, God can introduce a drought. God does not need to script the resulting protest, political fallout, business failures, family arguments or migration. Those should emerge from the world, institutions and characters.

20. Engine Interaction

> *GOD → WORLD CONDITIONS → WORLD ENGINE → CONSTRAINT PRESSURE → SIMULATION ENGINE → CHARACTER ENGINE → LLM REASONING → CHARACTER ACTION → SIMULATION ENGINE → WORLD STATE → NEXT TICK*

The LLM sits inside the causal loop; it does not replace the causal loop.

The world establishes reality. The character responds. The response changes reality. The changed reality affects future decisions.

21. Example: A Drought

Initial world: A town has a reservoir, population, farms, businesses, households and government.

World physics: Rainfall becomes unusually low while consumption continues. Reservoir level falls.

Constraint pressure: Water stress rises before the reservoir is empty.

Simulation: Prices, conservation, government action, agriculture and business conditions may change.

LLM-mediated response: Characters reason about their own circumstances. One stockpiles water; another trusts the government; another protests; a politician reassures people; a shopkeeper changes prices.

Emergent consequence: Those actions change relationships, finances, public opinion, health, political pressure and resource availability.

The player may experience the drought directly, indirectly, or not at all.

22. Hard Rules

• The player is never the center of all world activity.

• The LLM cannot arbitrarily overwrite authoritative world facts.

• Events are not created solely to make the player's story interesting.

• Hidden information is not automatically exposed to the player.

• Characters and resources cannot violate physical/world constraints.

• Every NPC should be treated as a person with at least a coherent history and state.

• The player is not required to trigger meaningful events.

• Not every consequence becomes a quest.

• Randomness should not replace causality where world conditions can explain an outcome.

23. V1 Scope

The first engine prototype should be deliberately small. Its purpose is to prove that the world can live before significant effort is spent on graphical UI.

• Small town / contained region

• Manageable population

• One player

• Tick-based time

• Locations and physical presence

• Basic resources and constraints

• Basic health

• Basic money and employment

• Relationships

• Knowledge and memory

• Autonomous character activity

• Emergent, scheduled and God events

• LLM-assisted reasoning and interaction

• God/developer inspection mode

• Terminal interface as the first testing interface

The terminal is a laboratory and control surface, not the final game UI. The simulation must remain independent of presentation.

24. North Star

Build the world first. Build the game interface second.

The first major milestone is a world that can advance through time and continue doing meaningful things without the player.

If the player leaves and returns to find relationships changed, people made decisions, resources moved, events occurred and circumstances evolved, then Moh Maya has begun to become the world promised by the Bible.

> *Does the player believe that they actually lived?*

Every engine exists to serve that experience. The World Engine establishes reality. The Simulation Engine makes reality change. Character and LLM layers create human behavior. Relationships, Knowledge, Health, Economy, Government and Events create the interconnected pressures of life. The God Layer introduces controlled disturbances without dictating every consequence.

Together they should produce a world that is not waiting for the player.

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